Plotluck

AI that helps London's small restaurants choose where to open, using what made past restaurants survive, and points new owners toward cuisines their neighbourhoods are missing.

Initial concept, details subject to change · Team of 5 · theme: Wildcard: Beyond the Obvious → Community Impact

1. The problem

COVID closed a large share of London's independent restaurants, including the restaurant one of our teammates' father ran. People reopening or starting out still choose a site on gut feel: they walk the street, look at the rivals nearby and sign a lease that is often the biggest financial risk they will ever take. Chains have location analysts; small owners don't.

At the same time, London is one of the most diverse cities in the world, but many food cultures are under-represented. Communities and international students often can't find food from home nearby, even where there are clearly enough people to support it. That unmet demand is invisible to someone choosing a site.

2. What we're building

A web app where a would-be owner describes the restaurant they want to open: **cuisine** first, plus other properties such as price band, format and size (final set to be confirmed in the build). The app then:

- scores **every London postcode district** (e.g. E8, SE15) from 0 to 1 for that restaurant;
- shows results as **colour-coded nodes** filtered by a user-set **threshold** (default 0.6). Zoomed out, big nodes summarise an area; zooming in splits them into districts, so owners see a range of good options, not one 'answer';
- shows a **driver chart** for each district, explaining which factors push its score up or down.

It deliberately does not plot existing competitors. Owners already over-weight 'who is nearby'; we want them to see survival evidence and unmet demand instead.

3. What makes it 'beyond the obvious'

- **Learns from closures, not just successes.** Most site tools use footfall and rent. We label thousands of past London restaurants by how long they lasted, using business dissolutions and hygiene-register removals as the signal of failure.
- **Finds cultural demand gaps.** Census data on ethnicity, country of birth, language and students is compared with how many restaurants of a cuisine already exist, so the tool can flag where a community is under-served.
- **Built for the smallest players.** Free, explainable and open data only. No paid footfall feeds.

4. Data

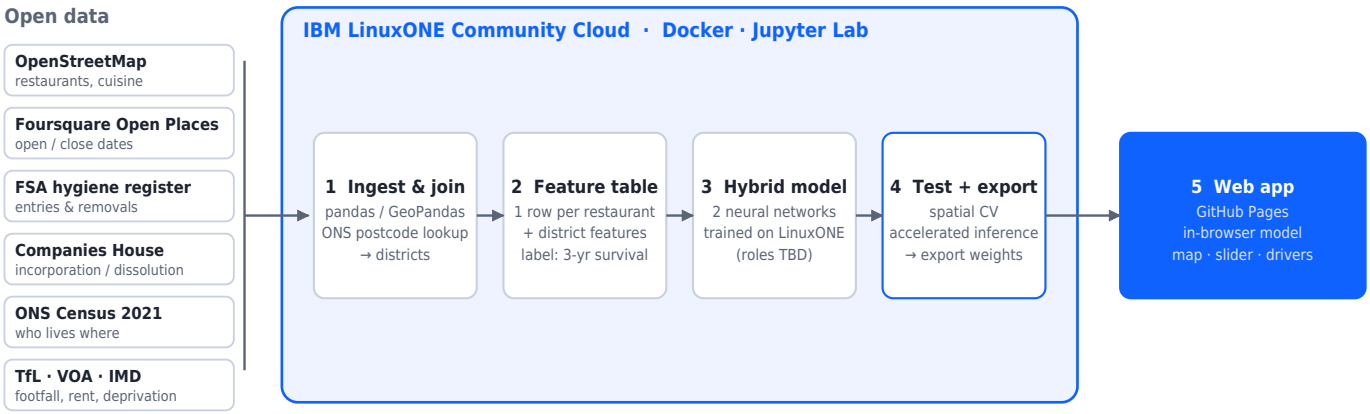
Dataset	Used for
OpenStreetMap	Restaurant list, location, cuisine, attributes
Foursquare Open Places	Categories; opening and closing dates
FSA hygiene register	When food businesses appear / disappear
Companies House	Incorporation and dissolution → longevity
ONS Census 2021	Population, age, ethnicity, country of birth, language, students
TfL station entries/exits	Footfall proxy
VOA rateable values	Rent and cost proxy
Index of Multiple Deprivation	Local income and deprivation
ONS Postcode Directory	Maps every record to a postcode district

All sources are free and open, and all are aggregate or business-level. **No personal data** is used.

5. Modelling approach

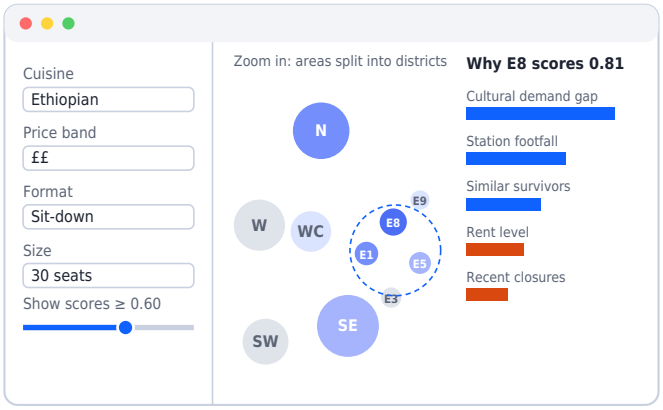
- **Training unit: one row per restaurant.** London has only ~300 postcode districts, far too few rows for a model to learn from. Instead each existing restaurant becomes a row: its own properties plus its district's features.
- **Label: survived 3+ years** (binary). Restaurants opened less than 3 years ago are excluded, because their outcome isn't known yet. The exact definition will be refined as we match records across sources.
- **Models:** a **hybrid of two neural networks** built with open-source libraries; their exact roles and architectures will be chosen during the build. Validated with **spatial cross-validation** (whole districts held out) so the score isn't flattered by neighbouring restaurants leaking into training.
- **Scoring:** for the user's chosen properties we create a hypothetical restaurant in every district, score it, and return all districts above the threshold. Trained weights are exported (format TBD, e.g. ONNX) and run **in the browser**, so the site stays live after the event (fallback: precomputed scores).
- **Explanations:** per-feature contributions (method TBD) feed a clear, visual driver chart, so every score can be explained without an LLM.

6. Architecture



Open data is joined to postcode districts, turned into a restaurant-level training table, and used to train two neural networks on LinuxONE. The evaluated weights are exported and run in the browser on a static GitHub Pages site.

7. User experience



Mock-up: the owner sets their restaurant's properties and threshold; acceptable areas show as colour-coded nodes that split into districts on zoom, and clicking one shows what drives its score.

8. Why IBM LinuxONE

- **All AI on IBM Z.** Ingest, training of both networks and evaluation run in Jupyter Lab on LinuxONE (s390x), using open-source PyTorch / TensorFlow / scikit-learn.
- **On-chip inference.** We benchmark inference on the IBM Z Integrated Accelerator for AI before exporting the weights.
- **Containerised.** The workload runs in Docker on the Community Cloud VM. Every step, from login to training to export, is evidenced with screenshots.
- **Secure by default.** Data stays inside an isolated enterprise environment, which matters if councils or lenders later add their own non-public data.

9. Responsible AI

- Scores are **decision support**, shown as a range of options with reasons, never a single verdict.
- Survival is shaped by past conditions, including COVID. We'll check how scores shift when the pandemic years are excluded and report it.
- The cultural demand feature uses only aggregate Census counts and is designed to **widen** choice, not to steer cuisines only toward 'their' communities.
- Record matching between sources is imperfect; match rates will be reported alongside model accuracy.

10. Delivery plan & stretch goals

Phase	Output
Data	Joined restaurant table with longevity label; district feature table
Models	Hybrid of two neural nets; spatial CV results; driver explanations
Platform	Training + accelerated inference on LinuxONE; weight export; screenshots
App	GitHub Pages site with in-browser scoring; 3-min demo video

Stretch: a true survival model (time-to-closure) instead of a 3-year cut-off; clustering districts into 'area types' as an extra model input; a live LinuxONE API during judging; extending beyond London.

11. Team workflow (one shared server)

- GitHub issue → local branch → test on sample data → PR → merge to main → server operator pulls and runs on the full data.
- One team member operates the server at a time. Linux-only errors are hotfixed on the server, pushed to a fix/linux-* branch and merged through a PR.
- An evidence checklist in the repo tracks every LinuxONE screenshot needed for the video.